

NEXTAR QUANTUM SYSTEMS

Internship Capstone Project

Official Guidelines & Submission Requirements

Submission Date	30 th September, 2026
Status	Mandatory — prerequisite for Certificate of Achievement and Letter of Completion
Tracks	AI Engineering · Software Engineering · Brand & Strategy · Network Engineering
Submission Email	nextarquantumsystems@gmail.com

Overview

Every intern at Nextar Quantum Systems is required to complete a capstone project at the end of their internship. This capstone is compulsory: it is the final checkpoint used to assess practical understanding of the track you trained in, and completing it — with an approved submission — is a prerequisite for receiving your Certificate of Achievement and Letter of Completion. Interns who do not submit a capstone project, or whose submission does not meet the minimum requirements below, will not be issued these documents.

There are four tracks, one per section of interns. Each intern should only complete the capstone project for the track they were trained in. Read your track's section carefully, including the submission method, before you begin.

General Rules

1. Work must be original. Copied or plagiarized submissions (code, brand guides, or reports) will be disqualified.
2. Every submission — regardless of track — must include your full name, intern ID (check your Identity Card), track name, and cohort/batch in the email subject line or the submitted document.
3. Keep a copy of everything you submit until your certificate and letter of completion have been issued.
4. If you have questions about your task, reach out to your track supervisor before the deadline, not after.

Submission Methods — Quick Reference

There are two ways you may be asked to submit, depending on your track:

- GitHub + Email notification — used by AI Engineering and Software Engineering, since these tracks produce code.
- Direct Email Submission — used by Brand & Strategy and Network Engineering, since these tracks produce documents/reports rather than code repositories.

Full instructions are repeated at the end of each track section below so you don't need to flip back and forth.

SECTION 1

AI Engineering

Basic EDA · Retrieval-Augmented Generation (RAG) · YOLO Object Detection

Objective

Demonstrate your ability to explore and prepare a dataset, build a working Retrieval-Augmented Generation pipeline, and implement an object detection system using YOLO.

Tasks

Complete all three of the following mini-projects. Each should live in its own clearly labelled folder inside a single GitHub repository.

1. Exploratory Data Analysis (EDA): Choose any public dataset (Kaggle, UCI, or one assigned by your supervisor). Clean the data, handle missing values/outliers, and produce at least 5 meaningful visualizations with written insights explaining what each chart shows.
2. Retrieval-Augmented Generation (RAG): Build a simple RAG pipeline — ingest a small document set, chunk and embed it, store it in a vector store, and connect it to an LLM so it can answer questions grounded in your documents. Include at least 5 sample question/answer interactions showing it working.
3. YOLO Object Detection: Use a pretrained or fine-tuned YOLO model (YOLOv5/v8 or later) to detect objects in a set of images or a short video. Show sample output images with bounding boxes and a short note on the model's accuracy/limitations.

Deliverables

- A single GitHub repository with three folders: /eda, /rag, /yolo
- A top-level README.md summarizing what each folder contains and how to run it
- Notebooks or scripts, plus sample outputs (charts, Q&A logs, detection images) committed to the repo

Evaluation Criteria

- Correctness and functionality of each mini-project
- Quality and clarity of insights/documentation
- Code organization and reproducibility (can a supervisor clone and run it?)

HOW TO SUBMIT

1. Create a public (or private-with-access) GitHub repository named: nextar-capstone-ai-<yourname>.
2. Push all your code, notebooks, and sample outputs, along with the README.md described above.
3. Add nextarquantumsystems@gmail.com as a collaborator on the repository (Settings → Collaborators) if it is private.
4. Send a confirmation email to nextarquantumsystems@gmail.com with the subject line: "AI Engineering Capstone – [Your Full Name]", including your GitHub repository link in the body.
5. You will receive an acknowledgment once your submission has been received and reviewed.

SECTION 2

Software Engineering

Pure Python Attendance Management System · Django Implementation with UI

Objective

Build the same core system — an Attendance Management System — twice: once in pure Python (no web framework), and once as a Django web application with a working user interface.

Tasks

1. Pure Python Version: Build a command-line Attendance Management System using only core Python (no Django/Flask). It should support: adding students/employees, marking daily attendance, viewing attendance records by date or by person, and generating a simple summary report (e.g., % attendance per person). Data can be stored in a file or a local database (e.g., SQLite via sqlite3).
2. Django Version: Rebuild the same core functionality as a Django web application with a working UI (HTML templates, or Django admin plus custom views — your choice, but there must be a usable interface, not just the admin panel). It should support the same core features: register people, mark attendance, and view/filter attendance records.

Deliverables

- A single GitHub repository with two folders: /python-version and /django-version
- A top-level README.md with setup/run instructions for both versions
- requirements.txt for the Django version, and any sample data/screenshots showing the UI in action

Evaluation Criteria

- Both versions work as described and cover the same core features
- Clean, readable code with sensible structure (functions/classes, Django apps/models/views)
- UI is functional and usable in the Django version
- Clear setup instructions that actually work when followed

HOW TO SUBMIT

1. Create a public (or private-with-access) GitHub repository named: nextar-capstone-swe-<yourname>.
2. Push both the pure Python version and the Django version, each in its own folder, with the README.md and requirements.txt described above.
3. Add nextarquantumsystems@gmail.com as a collaborator on the repository (Settings → Collaborators) if it is private.
4. Send a confirmation email to nextarquantumsystems@gmail.com with the subject line: "Software Engineering Capstone – [Your Full Name]", including your GitHub repository link in the body.
5. You will receive an acknowledgment once your submission has been received and reviewed.

SECTION 3

Brand & Strategy

Develop a Complete Brand Guide

Objective

Demonstrate your ability to think strategically about a brand and translate that thinking into a clear, usable brand guide — the kind a company could actually hand to a designer or marketer.

Tasks

Choose a brand: either a real brand you'll rebrand/refresh, or a fictional brand/business you invent for this exercise (subject to your supervisor's approval). Develop a complete brand guide covering:

- Brand overview: mission, vision, values, and positioning statement
- Target audience: at least one clear audience persona
- Brand personality and tone of voice, with do's and don'ts
- Logo usage guidelines (clear space, minimum size, incorrect usage examples)
- Color palette with exact codes (HEX/RGB/CMYK)
- Typography (primary and secondary fonts, usage hierarchy)
- Imagery/photography style direction
- At least 2 sample applications (e.g., business card, social media post, letterhead, or app screen) showing the brand in use

Deliverables

- One polished document (PDF or PowerPoint/Canva export) presenting the full brand guide
- Design software is your choice — Canva, Figma, Adobe, PowerPoint — as long as the final export is a clean, professional PDF (or PPT)

Evaluation Criteria

- Strategic thinking behind the brand direction, not just visual polish
- Completeness against the checklist above
- Consistency between the guidelines and the sample applications shown
- Overall presentation quality

HOW TO SUBMIT

1. Export your finished brand guide as a single PDF (preferred) or PPTX file. Name the file: Brand Guide – [Your Full Name].pdf
2. Attach it directly to an email addressed to nextarquantumsystems@gmail.com.
3. Use the subject line: "Brand & Strategy Capstone – [Your Full Name]".
4. In the email body, include a 2–3 sentence summary of the brand you chose and why.
5. No GitHub repository is required for this track — submit by email only.

SECTION 4

Network Engineering

Subnetting · VLSM · General Networking Knowledge

Objective

Demonstrate practical command of IP addressing and subnet design, and show a broader working knowledge of core networking concepts.

Tasks

1. Subnetting: Given a base network (your supervisor will assign one, e.g., 192.168.10.0/24), divide it into the number of subnets specified in your brief. For each subnet, show: network address, broadcast address, usable host range, and subnet mask (in both dotted-decimal and CIDR notation). Show your working, not just final answers.
2. VLSM (Variable Length Subnet Masking): Given a scenario with departments/sites requiring different numbers of hosts (e.g., Sales: 50 hosts, IT: 20 hosts, Guest Wi-Fi: 10 hosts), design an efficient VLSM addressing scheme for the whole network with minimal wasted address space. Present it as a clear addressing table.
3. General Networking Knowledge: Answer the set of short-answer/theory questions provided by your supervisor, covering topics such as the OSI/TCP-IP models, common protocols, routing vs switching basics, and troubleshooting scenarios.

Deliverables

- One document (Word or PDF) containing all three sections above, clearly labelled
- Working/calculations shown for subnetting and VLSM, not just final tables
- Optional but encouraged: a simple topology diagram (hand-drawn/scanned, Packet Tracer, or any diagramming tool) showing how your VLSM design maps onto a network

Evaluation Criteria

- Accuracy of subnetting and VLSM calculations
- Efficient use of address space in the VLSM design
- Depth and accuracy of answers in the general knowledge section
- Clarity of presentation (tables, labelling, working shown)

HOW TO SUBMIT

1. Export your completed work as a single PDF (preferred) or Word document. Name the file: Network Capstone – [Your Full Name].pdf
2. Attach it directly to an email addressed to nextarquantumsystems@gmail.com.
3. Use the subject line: "Network Engineering Capstone – [Your Full Name]".
4. If you created a topology diagram in a separate file, attach it as well (or include it as an extra page in the same PDF).
5. No GitHub repository is required for this track — submit by email only.

Final Note

Once your submission has been reviewed and approved by your track supervisor, your Certificate of Achievement and Letter of Completion will be processed. If any part of your submission needs revision, you will be contacted at the email address you submitted from — so please use an email you check regularly and double-check your GitHub links and attachments before sending.